

OptiCrop – Smart Agricultural Production Optimization Engine

This documentation is adapted for the Smart Agriculture project. It replaces the sample CaptionAI project with an AI-powered crop recommendation system that predicts suitable crops using soil and environmental parameters. ■filecite■turn19file0■

Project Description

OptiCrop uses Machine Learning to recommend the most suitable crop based on soil nutrients (N, P, K), temperature, humidity, pH, and rainfall. The system is built using Python, Flask, HTML, CSS, and a trained ML model (crop_model.pkl).

Technical Architecture

Frontend: HTML/CSS

Backend: Flask (Python)

ML Model: Trained crop recommendation model (.pkl)

Dataset: Crop recommendation dataset

Deployment: Local Flask server / optional cloud deployment.

Project Workflow

1. Requirement Analysis
2. Data Collection & Preprocessing
3. Model Training
4. Flask Integration
5. Frontend Development
6. Testing & Validation
7. Documentation & Deployment

Run Instructions

1. Clone the repository.
2. Install dependencies: `pip install -r requirements.txt`
3. Run: `python app.py`
4. Open the local URL displayed in the terminal.

Conclusion

OptiCrop demonstrates how Artificial Intelligence and Machine Learning can assist farmers in making informed crop selection decisions, improving productivity and supporting smart agriculture.